

## Slope and Deflection of Beams

There are a number of different approaches used to determine the slope and deflection of beams. These techniques are used to conduct design for stiffness and to optimise structures for minimum weight and structural efficiency. We are going to consider two widely used methods, namely Superposition and Strain Energy.

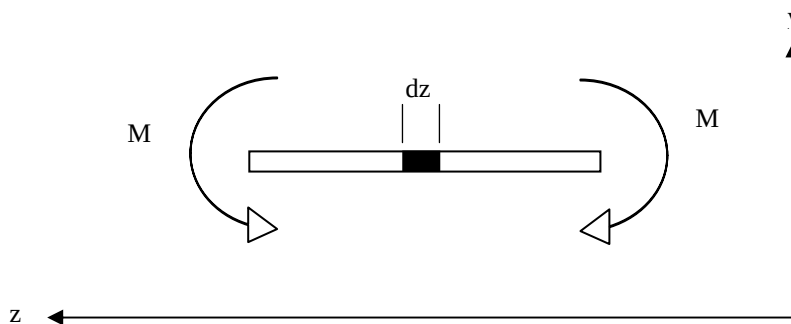
The determination of the slope and deflection of beams is equally as important as the study of stresses due to bending. Indeed, design criteria often focus on these two separate aspects of a component.

The total deflection of a beam is due to a very large extent to the deflection caused by bending, and to a very much smaller extent to the deflection caused by shear. The part of the deflection caused by shear will not be dealt with in this section.

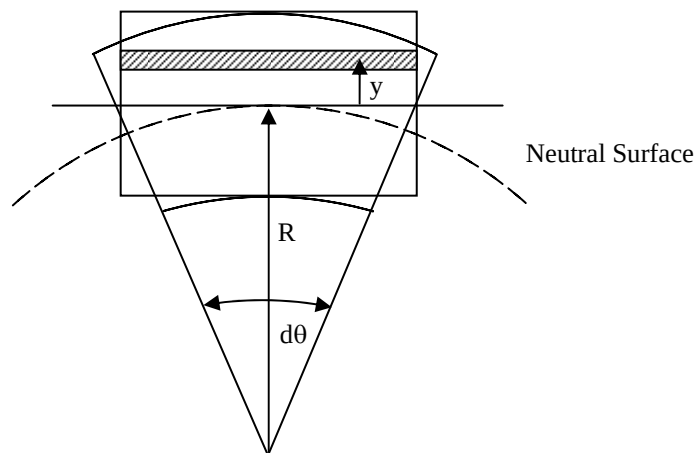
In practice it is usual to put a limit on the allowable deflection, in addition to the stresses, although in some cases allowable deflection can be quite large.

It is important that we can calculate the deflection of a beam of given section, since for the given conditions of span and load it would be possible to adopt a section which would meet a strength failure criterion but would give an unacceptable deflection.

Consider a small length  $dz$  of the beam shown:



This length, under the action of pure bending, will bend into a circular arc, the radius being  $R$ .



The neutral surface shown is unstressed and will remain the same length. The angle subtended by the element when stressed is  $d\theta$ .

Consider the stress acting on a 'fibre' of the material (shaded), located a distance  $y$  above the neutral surface, assuming a symmetrical section and only a bending moment acting in the  $M_x$  sense. The stress on the fibre is then given by:

$$\sigma_z = \frac{M_x \cdot y}{I_{xx}}$$

Because the loading is pure bending,  $\sigma_z$  is the only stress acting and is therefore directly related to Young's Modulus:

$$E = \frac{\sigma_z}{\epsilon_z}$$

Combining these two equations:

$$\epsilon_z = \frac{M_x \cdot y}{E \cdot I_{xx}}$$

The axial strain may be approached by considering the change in length of the element:

$$\varepsilon_z = \frac{\text{change in length of element}}{\text{original length}}$$

Original length of element = length of neutral axis =  $R \cdot d\theta$

New length of element =  $(R+y) d\theta$

$$\varepsilon_z = \frac{(R+y)d\theta - Rd\theta}{Rd\theta}$$

Therefore,

$$\varepsilon_z = \frac{y}{R}$$

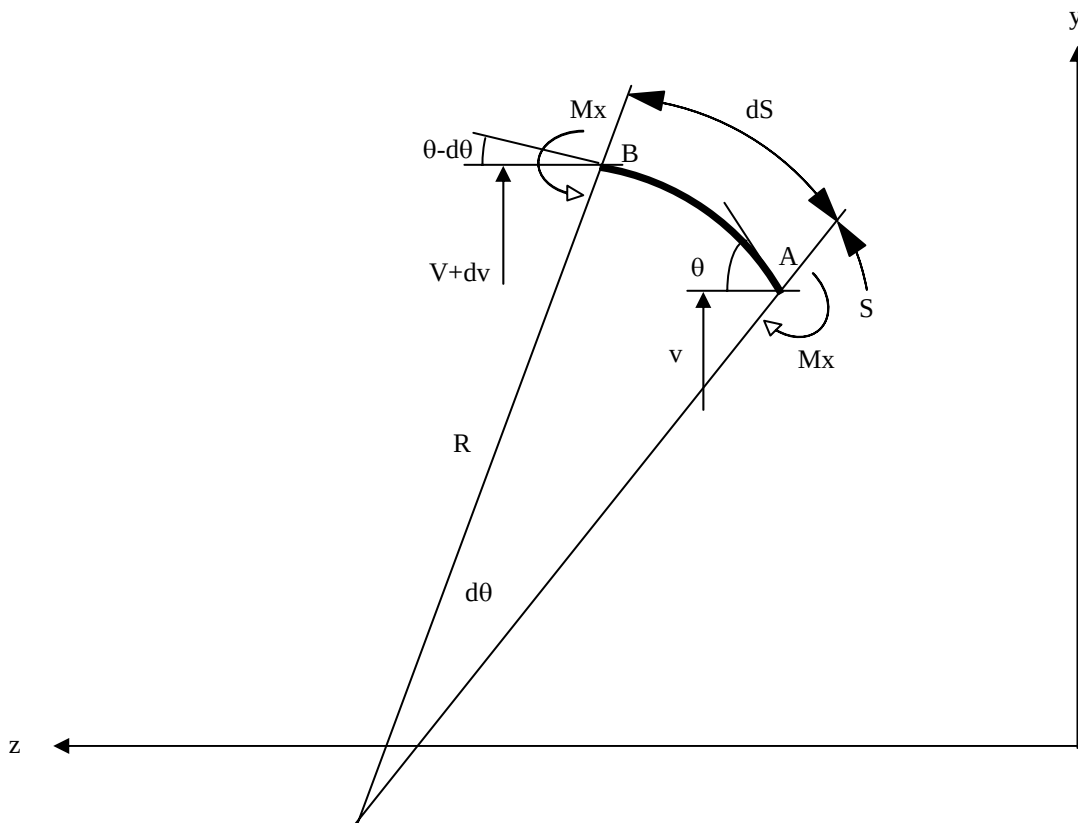
Equating the two expressions for strain, gives,

$$\frac{M_x \cdot y}{E \cdot I_{xx}} = \frac{y}{R}$$

Therefore,

$$\boxed{\frac{M_x}{E \cdot I_{xx}} = \frac{1}{R}}$$

Now consider a length of beam  $ds$ , as shown.



It is usual to describe deflection in the three co-ordinate directions, by symbols  $u$ ,  $v$  and  $w$  respectively.

Thus the point A has a deflection  $v$  in the  $y$  direction, and the point B a deflection of  $v + dv$ .

The tangent at A makes an angle of  $\theta$  with the  $z$  axis and the tangent at B an angle of  $\theta - d\theta$ , this decrease in slope of  $d\theta$  over a length change from  $s$  to  $s + ds$  corresponds to a positive  $M_x$ , as shown.

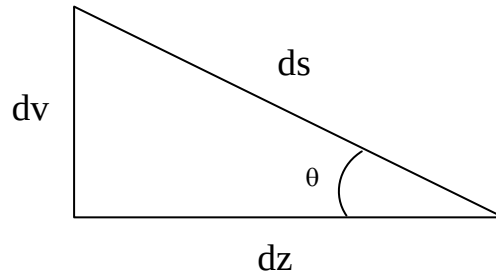
The analysis will be restricted to beams in which deflections are always small (not greater than the cross section dimensions). This means that by correct orientation of the axis system and the beam, the angle  $\theta$  will always be small.

Taking out the basic triangle formed by the beam AB:

As  $\theta$  is very small

$$\cos \theta \approx 1 = \frac{dz}{ds}$$

$$\therefore dz = ds$$



Also:

$$\tan \theta \approx \theta = \frac{dv}{dz}$$

Using trigonometry, and considering the diagram on the previous page, we can write:

$$ds = R.d\theta \quad \text{or} \quad \frac{d\theta}{ds} = \frac{1}{R}$$

But the slope decreases as 's' increases, so we can write:

$$\frac{d\theta}{ds} = -\frac{1}{R}$$

$$\therefore \frac{1}{R} = -\frac{d\theta}{ds} = -\frac{d\theta}{dz} = -\frac{d^2v}{dz^2}$$

$$\boxed{\therefore \frac{1}{R} = -\frac{d^2v}{dz^2}}$$

Equating the two expressions for curvature gives:

$$\frac{1}{R} = \frac{M_x}{E.I_{xx}} = \frac{-d^2v}{dz^2}$$

Note that bending could have taken place about the y axis:

$$\frac{M_y}{E.I_{yy}} = \frac{-d^2u}{dz^2}$$

These equations are only true if one or both of the x and y axis are 'Principal Axes'.

What follows will relate the first of these two equations, but by a change of suffixes the equations could be changed to fit deflection in the 'x' direction.

It may of course be necessary, depending on the loading, to calculate deflection in both directions and then take the vector sum of the two components to obtain the maximum TOTAL deflection.

## The Principle of Superposition

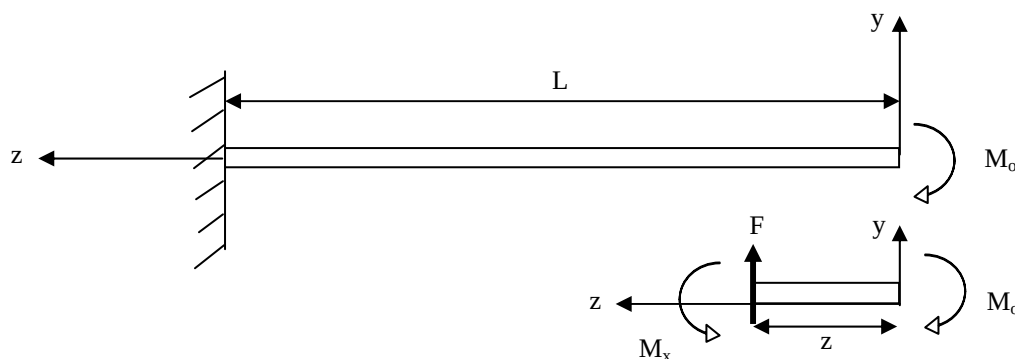
These equations can be applied to three simple cases of beams with bending moment  $M_x$  as a continuous function along the length of the beam, thus slopes and deflections may be obtained at the ends of the beams. These cases form the basis of a practical beam deflection techniques known as 'Superposition'.

These three beam cases are:

- 1 Beam with an end moment
- 2 Beam with an end point load
- 3 Beam with a UDL

These beams are NOT in a state of pure bending, but the shear stiffness of most practical beam sections is so high that shear deflection can be ignored compared with bending deflection.

### CASE 1 – Beam with an end moment



Applying equilibrium to the small section of the beam

$$M_x - M_o = 0$$

$$\text{Therefore, } M_x = M_o$$

Hence applying the deflection equation:

$$\frac{M_o}{E.I_{xx}} = \frac{-d^2v}{dz^2}$$

From this we can obtain,

$$E.I_{xx} \cdot \frac{d^2 v}{dz^2} = -M_o$$

$$E.I_{xx} \cdot \frac{dv}{dz} = -M_o z + A$$

$$E.I_{xx} \cdot v = \frac{-M_o z^2}{2} + Az + B$$

Where A and B are constants of integration.

Now,  $\frac{dv}{dz} = 0$  when  $z=L$   $0 = -M_o L + A$  Therefore,  $A = M_o \cdot L$

$V=0$  when  $z=L$   $0 = \frac{-M_o L^2}{2} + M_o L^2 + B$  Therefore,  $B = \frac{-M_o L^2}{2}$

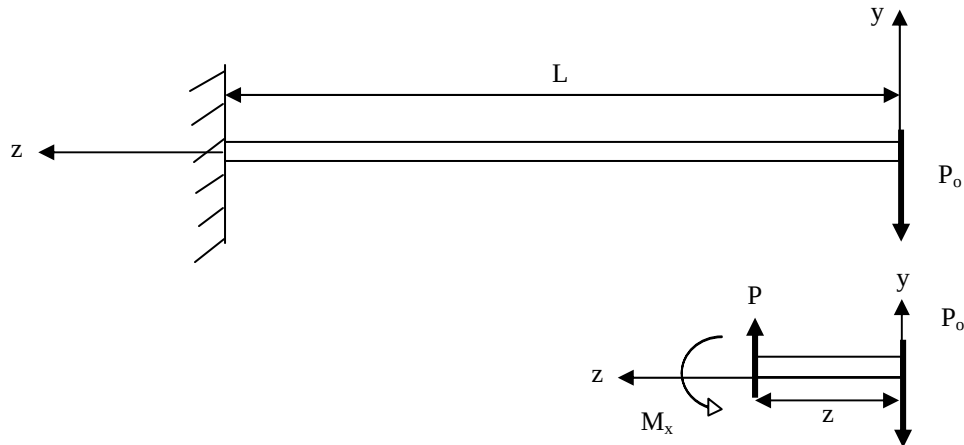
Hence,

Slope,  $\frac{dv}{dz} = \frac{1}{EI_{xx}} (-M_o Z + M_o L)$

Deflection,  $v = \frac{1}{EI_{xx}} \left( \frac{-M_o Z^2}{2} + M_o L \cdot Z - \frac{M_o L^2}{2} \right)$



## CASE 2 – Beam with an end load



Applying equilibrium to the small section of the beam:

$$M_x - P_o \cdot z = 0$$

Therefore,  $M_x = P_o \cdot z$

Hence applying the deflection equation,

$$\frac{P_o z}{EI_{xx}} = \frac{-d^2 v}{dz^2}$$

This leads to the following solution,

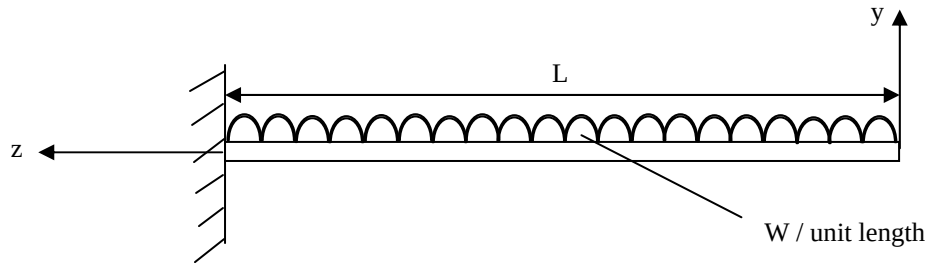
Slope,

$$\frac{dv}{dz} = \frac{1}{EI_{xx}} \left( \frac{-P_o z^2}{2} + \frac{P_o L^2}{2} \right)$$

Deflection,

$$v = \frac{1}{EI_{xx}} \left( \frac{-P_o z^3}{6} + \frac{P_o L^2 z}{2} - \frac{P_o L^3}{3} \right)$$

### CASE 3 – Beam with a uniformly distributed load



By a similar analysis to that above:

Slope,

$$\frac{dv}{dz} = \frac{1}{EI_{xx}} \left( \frac{-w.Z^3}{6} + \frac{w.L^3}{6} \right)$$

Deflection,

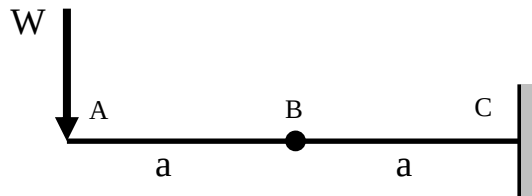
$$v = \frac{1}{EI_{xx}} \left( \frac{-wZ^4}{24} + \frac{wL^3Z}{6} - \frac{wL^4}{8} \right)$$

### Summary of Results

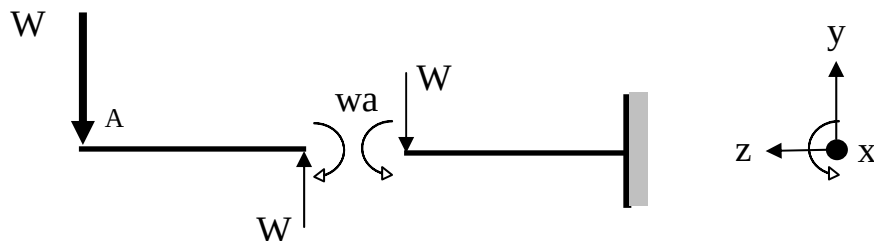
|  | Tip deflection              | Tip slope                |
|--|-----------------------------|--------------------------|
|  | $\frac{M_0 L^2}{2E.I_{xx}}$ | $\frac{M_0 L}{E.I_{xx}}$ |
|  | $\frac{P_0 L^3}{3E.I_{xx}}$ | $\frac{PL^2}{2E.I_{xx}}$ |
|  | $\frac{wL^4}{8E.I_{xx}}$    | $\frac{wL^3}{6E.I_{xx}}$ |

### Example 1

Determine the deflection at point B of the following beam:



STEP 1: Cut the beam at the point of interest, B. Draw Free Body Diagrams to determine the forces and moments, at this cut point.



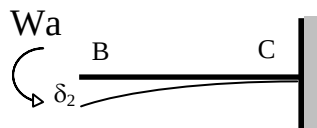
STEP 2: The deflection at B,  $\delta_B$ , has two components

$\delta_1$  due to W at B



$$\delta_1 = \frac{Wa^3}{3EI}$$

$\delta_2$  due to Wa at B



$$\delta_2 = \frac{Wa^3}{2EI}$$

Where:

$$\delta_B = \delta_1 + \delta_2$$

$$\delta_B = \frac{Wa^3}{3EI} + \frac{Wa^3}{2EI} = \frac{5Wa^3}{6EI}$$

It is interesting to note that the deflection at A,  $\delta_A$  is more than twice the deflection at B.

$$\delta_A = \frac{W(2a)^3}{3EI} = \frac{8Wa^3}{3EI}$$

Now considering the slopes at A and B:

$$\theta_A = \frac{W(2a)^2}{2EI} = \frac{2Wa^2}{EI}$$

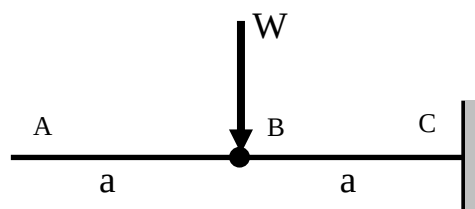
Slope at B is due to slope from W at B plus the slope from Wa at B:

$$\theta_B = \frac{Wa^2}{2EI} + \frac{Wa}{EI} \times a = \frac{3Wa^2}{2EI}$$

i.e. The slope at A is greater than the slope at B.

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Now consider the following beam.



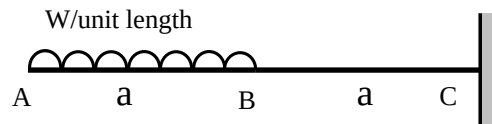
$$\delta_A = \frac{Wa^3}{3EI} + \frac{Wa^2}{2EI} \times a = \frac{5Wa^3}{6EI}$$

Maxwell's Reciprocal Rule: Deflection at A due to load at B is the same as the deflection at B if the load is applied at A.

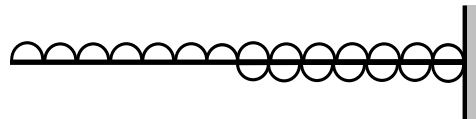
## Example 2

Determine the deflection at A.

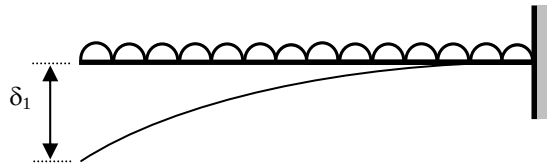
EI constant



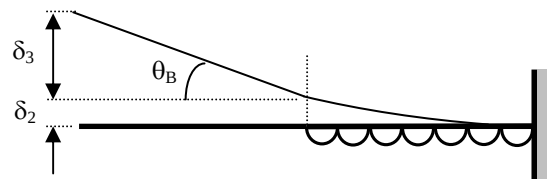
HINT: Split the load as:



$$\delta_1 = -\frac{W(2a)^4}{8EI}$$



$$\delta_2 = \frac{Wa^4}{8EI}$$



$$\delta_3 = \theta_B \times a = \frac{Wa^3}{6EI} \times a = \frac{Wa^4}{6EI}$$

The total deflection at A,  $\delta_A$ :

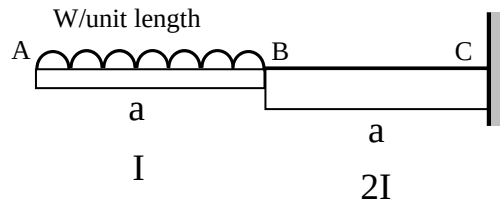
$$\delta_A = (\delta_2 + \delta_3) + \delta_1 = \frac{Wa^4}{8EI} + \frac{Wa^4}{6EI} - \frac{16Wa^4}{8EI}$$

$$\therefore \delta_A = -\frac{41}{24} \left[ \frac{Wa^4}{EI} \right]$$

Downwards

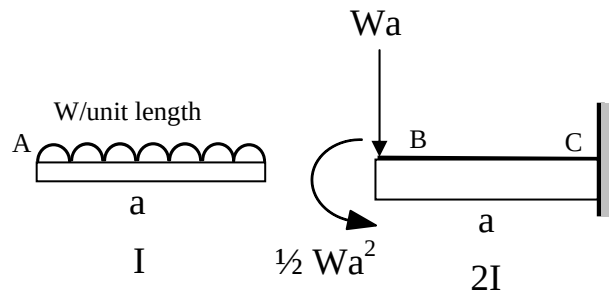
### Example 3

Determine the deflection at point A for the STEPPED beam shown.

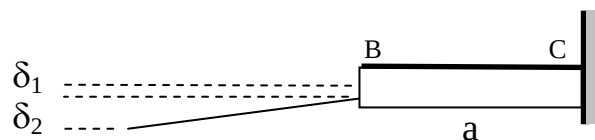


You can't use the simple method used previously as now the section properties are discontinuous.

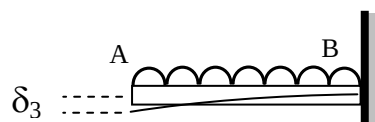
Consider section BC as a free body with constant cross-section of  $2I$  then add on the deflection due to AB.



Deflection at free end due to section BC only.



Then combine deflection due to AB.



Where:

$$\delta_1 = \delta \text{ due to } Wa + \delta \text{ due to } \frac{Wa^2}{2}$$

$$\delta_2 = [\theta \text{ due to } Wa + \theta \text{ due to } \frac{Wa^2}{2}] \times a$$

$$\delta_{\text{total}} = \delta_1 + \delta_2 + \delta_3$$

$$\delta_1 = \delta \text{ due to } Wa + \delta \text{ due to } \frac{1}{2} Wa^2$$

$$= \frac{Wa \times a^3}{3E(2I)} + \frac{\left(\frac{Wa^2}{2}\right)a^2}{2E(2I)}$$

$$= \frac{Wa^4}{6EI} + \frac{Wa^4}{8EI}$$

$$\therefore \delta_1 = \frac{7Wa^4}{24EI}$$



$$\delta_2 = \left[ \frac{Wa \times a^2}{2E(2I)} + \frac{\left(\frac{Wa^2}{2}\right)a}{E(2I)} \right] \times a$$

$$= \left[ \frac{Wa^3}{4EI} + \frac{Wa^3}{4EI} \right] \times a$$

$$\therefore \delta_2 = \frac{Wa^4}{2EI}$$



$$\delta_3 = \frac{Wa^4}{8EI}$$

$$\delta_{\text{total}} = \delta_1 + \delta_2 + \delta_3 = \frac{7Wa^4}{24EI} + \frac{12Wa^4}{24EI} + \frac{3Wa^4}{24EI}$$

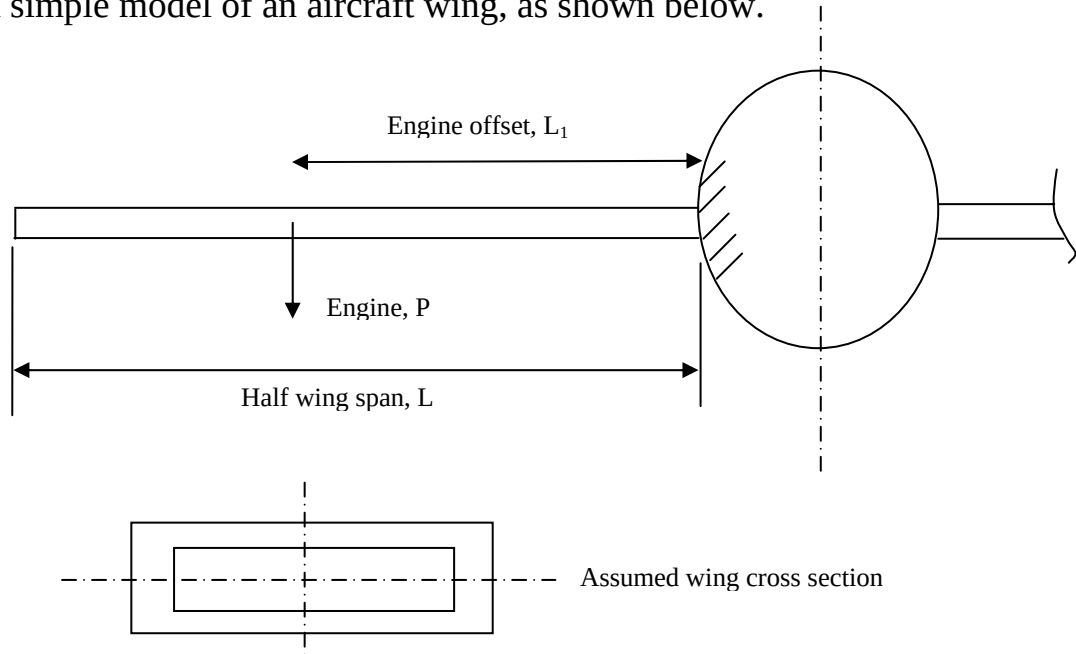
$$\delta_{\text{total}} = \frac{11}{12} \left( \frac{Wa^4}{EI} \right)$$



## Tutorials

### Question 1

Consider a simple model of an aircraft wing, as shown below.



The fuselage can be considered as a fixed end constraint.

Assume values for  $L$ ,  $L_1$ ,  $P$  and cross-sectional dimensions and then make an initial estimation of the wing tip deflection.

Assume the only significant load is the point load due to the engine.

Once completed, repeat your calculation adding in a continuous or discontinuous UDL, in addition to the engine load.

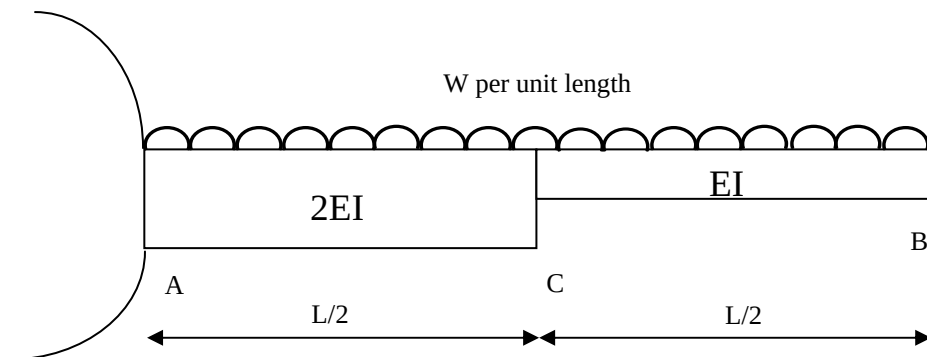
Calculate the new wing tip deflection.



**Question 2**

The wing of an aircraft can be modelled as a cantilever, with a step change in section, as shown below. The wing has a total length  $L$ . Section AC has a bending stiffness of  $2EI$  and section CB a bending stiffness of  $EI$ . Assuming the lift to be uniformly distributed along the wing, and equal to  $W$  per unit length, calculate:

1. the deflection and slope at the mid-span, C
2. the deflection at the wing tip, B



## Introduction to Strain Energy

Strain Energy is the single most important concept in stress analysis and it forms the basis of modern analytical methods.

The analysis of complex problems has been revolutionised by the development of matrix methods and in particular 'The Finite Element Method' – now accepted as THE method.

### A Simple Definition

When a body is deformed elastically it stores energy which it releases when the forces causing the deformation are relaxed.

or,

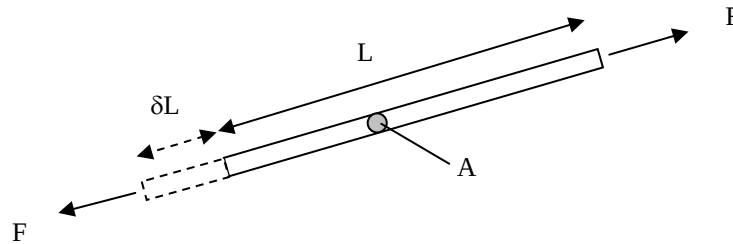
When a piece of material is deformed in simple bending, torsion, compression or tension etc. within its elastic range, work is done by the applied loading.

On removal of the loading the material returns to its undeformed state owing to the release of the stored ELASTIC STRAIN ENERGY (same magnitude as external work done).

Strain Energy can be used to solve real engineering problems in either a simple exchange of energy calculation or in a more complicated method called the **Unit Load Method** or **Method of Virtual Work**. More on this later!

## Strain Energy in Axial Members

Consider the simple case of AXIAL TENSION.



Where  $L$  is the original length,  $\delta L$  is the extension,  $A$  is the area and  $F$  is the applied force.

In order to develop an expression for Strain Energy we need to go back to basic elasticity.

We know:  $E = \frac{\sigma}{\varepsilon}$  and  $\sigma = \frac{F}{A}$   $\therefore E = \frac{F}{A\varepsilon}$

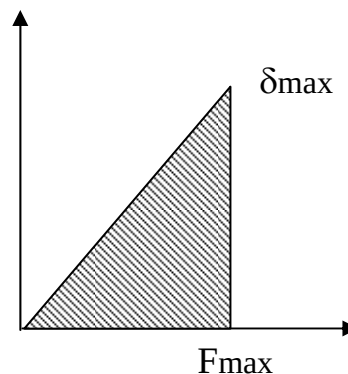
But,  $\varepsilon = \delta L/L$  hence  $E = \frac{FL}{A\delta L}$  or  $\delta L = \frac{FL}{AE}$

To find 'Energy Stored' we can consider 'External Work Done', WD.

WD = Area under graph =  $\frac{1}{2} F_{\max} \cdot \delta_{\max}$  Extension

But:  $\delta = \left( \frac{FL}{AE} \right)$

Therefore,  $WD = \frac{1}{2} F \left( \frac{FL}{AE} \right) = \frac{F^2 L}{2AE}$



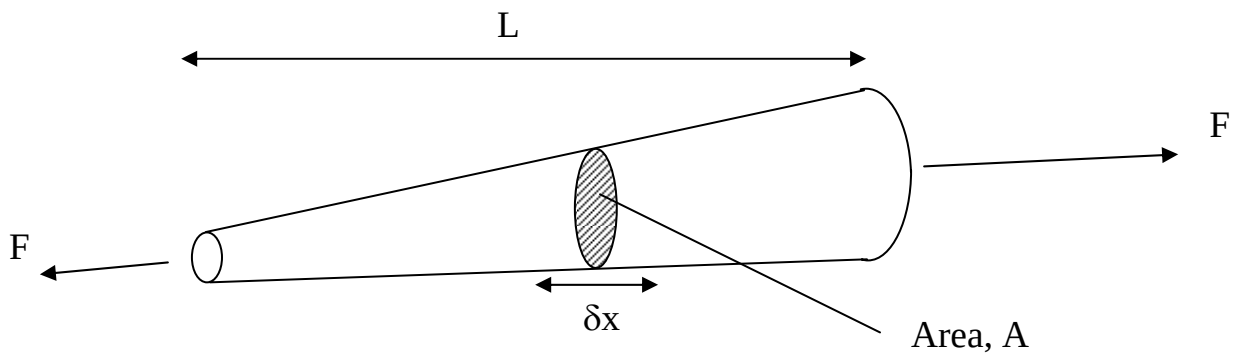
Since, Energy Stored Internally = Work Done

Therefore, Strain Energy,  $U$  in simple tension =  $\frac{F^2 L}{2AE}$

## Application to real axially loaded problems

In order to give us scope to deal with real engineering problems we must be able to deal with variable geometry.

Consider the following case:



Consider an element of length  $\delta x$  and Area,  $A$  where  $A$  is a function of  $x$  ie.  $A=f(x)$ .

From our previous result we know that the total energy in the member is given by:

$$U_{tot} = \int_0^L \frac{F^2}{2AE} dx$$

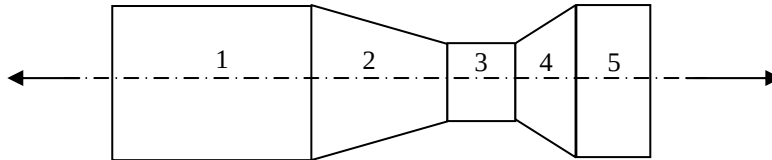
Since  $F$  is a constant and also  $E$ , in most cases,

$$U_{tot} = \frac{F^2}{2E} \int_0^L \frac{dx}{A}$$

Where  $A = f(x)$

We can now calculate the strain energy stored in a very complex rod by simply discretizing the component into a number of section of known form.

Consider the following example:



$$U_{\text{tot}} = U_1 + U_2 + U_3 + U_4 + U_5$$

This is also true in compression.

### Strain Energy in Bending and Torsion

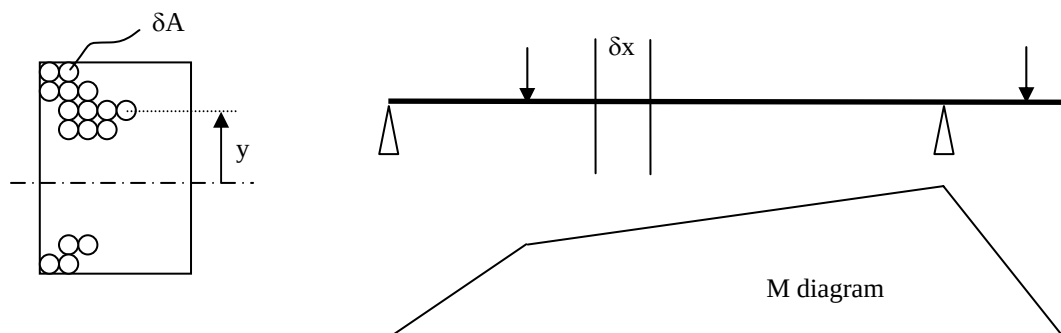
Of course not all structural components are axially loaded. Another common type of loading is bending and the associated structural element type is a beam.

A beam can be a simple uniform section member or a complex model of an aircraft wing.

Bending is a double problem, because:

1. Stress  $\sigma$  varies across the section
2. Bending moment  $M$  varies along the length

Viz:



Fibre Stress,

$$\sigma = \frac{M \cdot y}{I}$$

Fibre position

Each fibre behaves in Simple Tension or Compression. Therefore, we can use 'Axial Energy' for each individual fibre.

$$\text{Axial fibre force} = \sigma \cdot \delta A = \frac{My}{I} \cdot \delta A$$

To collect up all of the energy – firstly we must integrate across the section and then integrate along the length of the beam.

Consider a short length of beam,  $\delta x$

$$U = \int_0^L \int_{Bot}^{Top} \frac{F^2 dx}{2AE} = \int_0^L \int_{Bot}^{Top} \left( \frac{My \cdot dA}{I} \right)^2 \cdot \frac{1}{2AE} \cdot dx$$

$$U = \int_0^L \int_{Bot}^{Top} \frac{M^2}{I^2} (y^2 dA) \cdot \frac{dA \cdot dx}{2AE}$$

Fortunately for us we can write:

$$\int_{Bot}^{Top} y^2 dA = I \quad \text{and} \quad \int_{Bot}^{Top} dA = A$$

Therefore we end up with:

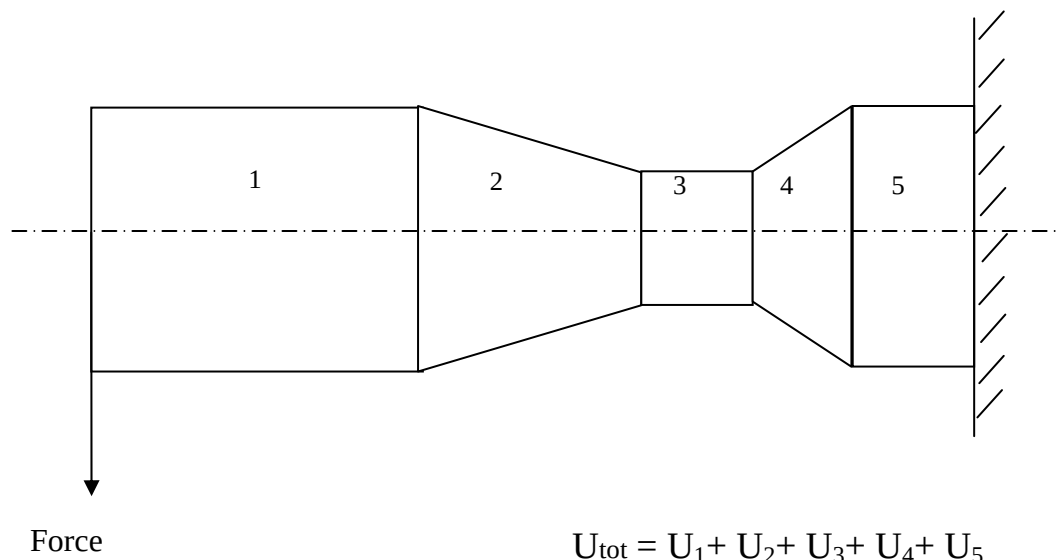
$$U_{Bending} = \int_0^L \frac{M^2}{2EI} dx$$

Where  $M=f(x)$  and  $I$  might be a function of  $x$ !

If ' $I$ ' is constant it can be brought outside the integral. If ' $I$ ' is a variable i.e. the beam has varying cross section (defined as function of  $x$ ), then  $I = f(x)$ .

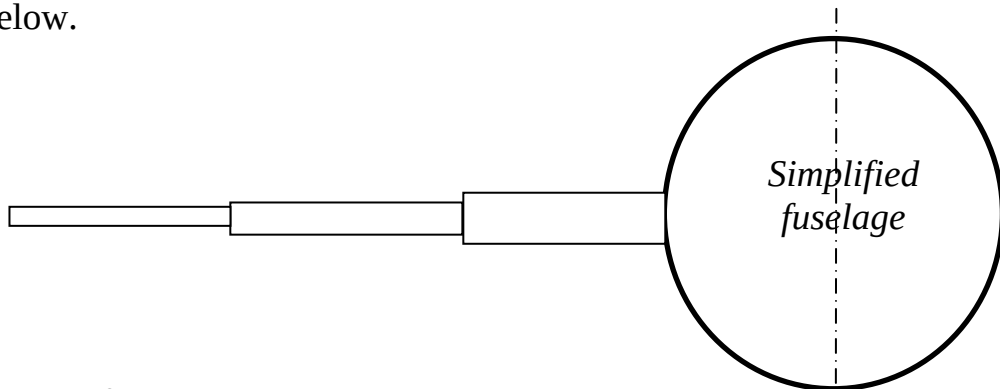
We can therefore find the 'Bending Energy' in very complicated beams of varying section along their length.

We can solve 'real' engineering problems by breaking down a complicated beam into regions – and then 'adding up' the respective energies.

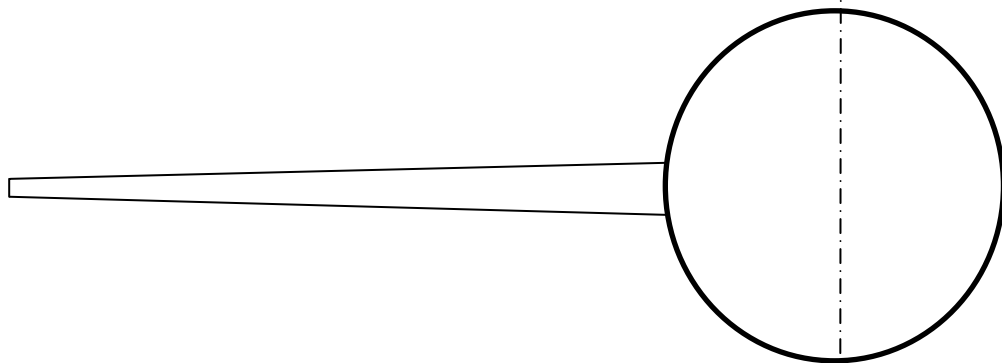


## Application to simplified wing models

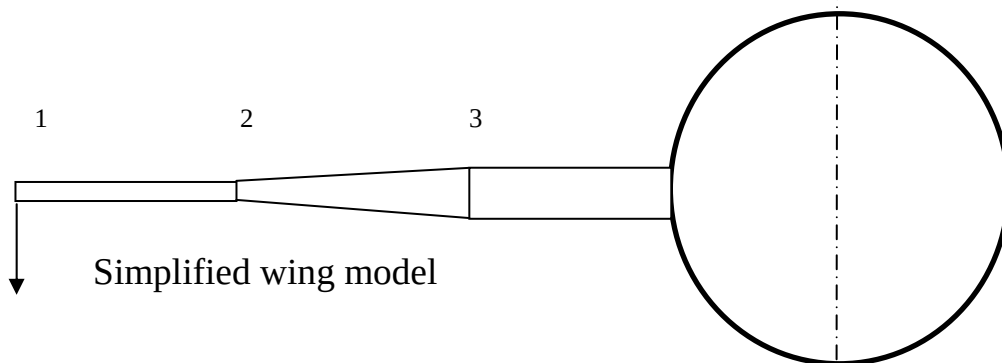
This approach can then be applied to a simplified stepped model of an aircraft wing, as shown below.



Or even a more refined linearly tapering wing model.



A further refinement could be to incorporate both types of geometry.



$$U_{\text{tot}} = U_1 + U_2 + U_3$$

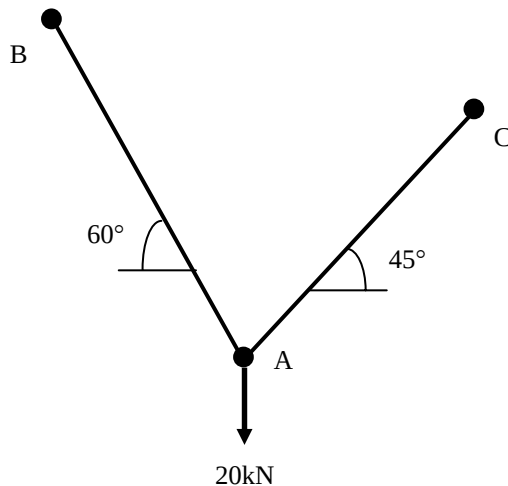


## Summary of Strain Energy Methods for Common Elements

| Element types        | Constant section                                         | Varying section                                                           |
|----------------------|----------------------------------------------------------|---------------------------------------------------------------------------|
| Axial Loading        | $U = \frac{F^2 L}{2 AE}$                                 | Where $A = f(x)$<br>$U = \frac{F^2}{2 E} \int_0^L \frac{dx}{A}$           |
| Bending              | Where $M = f(x)$<br>$U = \frac{1}{2 EI} \int_0^L M^2 dx$ | Where $M$ and $I = f(x)$<br>$U = \frac{1}{2 E} \int_0^L \frac{M^2}{I} dx$ |
| Shear due to torsion | $U = \frac{T^2 L}{2 JG}$                                 | Where $J = f(x)$<br>$U = \frac{T^2}{2 G} \int_0^L \frac{dx}{J}$           |

### Example 1

To introduce the technique, calculate the elastic strain energy stored in the simple pin jointed framework shown below.



Length AC = 0.45m

Length AB = 0.5m

Material: Steel,  $E = 208 \times 10^9 \text{ N/m}^2$

Both members are circular tube,  
50mm OD, 44mm ID

$CSA = 443 \times 10^{-6} \text{ m}^2$

$$U_{\text{tot}} = U_{AB} + U_{AC}$$

$$= \frac{F_{AB}^2 \times L_{AB}}{2AE} + \frac{F_{AC}^2 \times L_{AC}}{2AE}$$

To find forces we can use equilibrium.

$$\sum F_V = 0$$

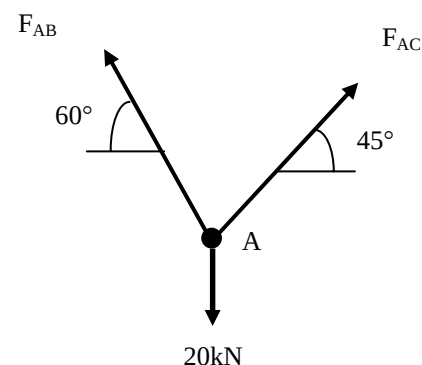
$$F_{AB} \sin 60 + F_{AC} \sin 45 - 20 \times 10^3 = 0$$

$$\sum F_H = 0$$

$$-F_{AB} \cos 60 + F_{AC} \cos 45 = 0$$

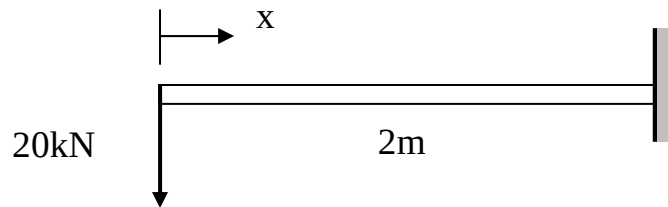
From which  $F_{AC} = 10.353 \times 10^3 \text{ N}$  and  $F_{AB} = 14.641 \times 10^3 \text{ N}$

$$\text{So, } U_{\text{TOT}} = \frac{(14.641 \times 10^3)^2 \times 0.5}{2.443 \times 10^{-6} \cdot 208 \times 10^9} + \frac{(10.353 \times 10^3)^2 \times 0.45}{2.443 \times 10^{-6} \cdot 208 \times 10^9} = 0.844 \text{ Nm}$$

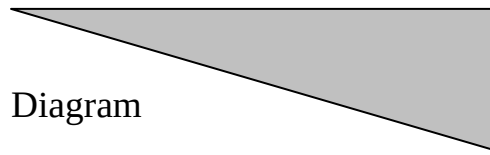


## Example 2

Now consider a simple box beam type problem. Again, calculate the total elastic strain energy stored.



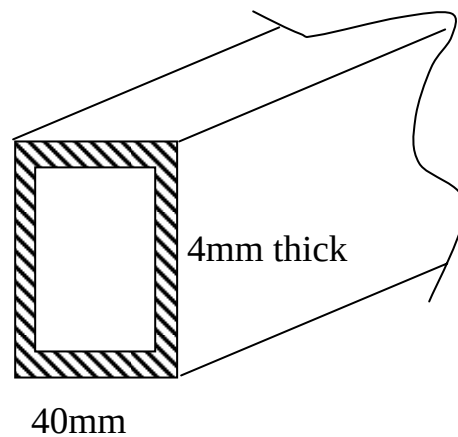
Bending Moment 'M' Diagram



Using:

$$U = \int_0^2 \frac{M^2}{2EI} dx$$

80mm



40mm

M is a function of x =  $20 \times 10^3 \cdot x$

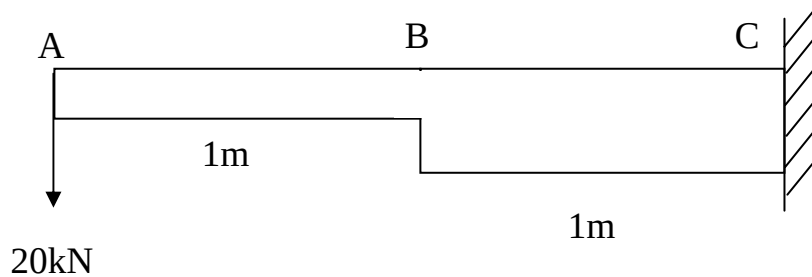
$$I = 0.711 \times 10^{-6} \text{m}^4$$

$$U = \frac{(20 \times 10^3)^2}{2 \times 70 \times 10^9 \times 0.711 \times 10^{-6}} \int_0^2 x^2 dx$$

$$\therefore \underline{U = 10.70 \times 10^3 \text{ Nm}}$$

### Example 3

Now consider a more complex beam problem, this time with a stepped beam. As before, calculate the total elastic strain energy stored.



Beam AB as for example 2, BC now twice the depth.

$$I_{AB} = 0.711 \times 10^{-6} \text{ m}^4$$

$$I_{BC} = \frac{0.04 \times 0.16^3}{12} - \frac{0.032 \times 0.152^3}{12} = 4.288 \times 10^{-6} \text{ m}^4$$

$$U_{TOT} = U_{AB} + U_{BC}$$

$$= \int_0^1 \frac{M_{AB}^2}{2EI_{AB}} dx + \int_1^2 \frac{M_{BC}^2}{2EI_{BC}} dx$$

$$\therefore U_{TOT} = \frac{(20 \times 10^3)^2}{2 \times 70 \times 10^9 \times 0.711 \times 10^{-6}} \int_0^1 x^2 dx + \frac{(20 \times 10^3)^2}{2 \times 70 \times 10^9 \times 4.288 \times 10^{-6}} \int_1^2 x^2 dx$$

$$= 4.018 \times 10^3 \left[ \frac{x^3}{3} \right]_0^1 + 0.666 \times 10^3 \left[ \frac{x^3}{3} \right]_1^2$$

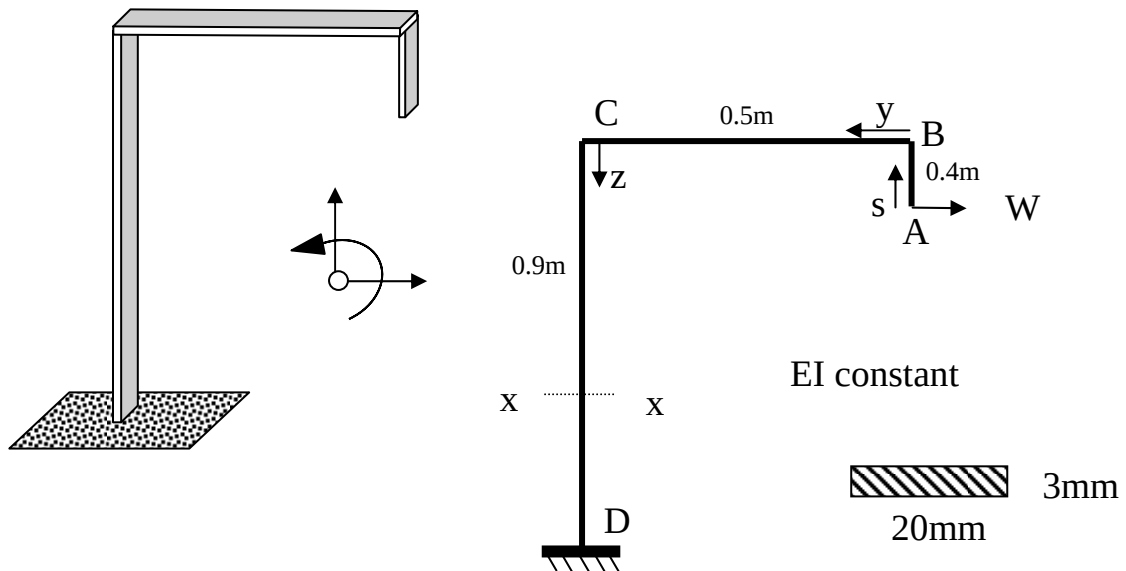
$$= 1.339 \times 10^3 + 1.554 \times 10^3 = 2.893 \times 10^3 \text{ Nm}$$

NOTE: Less energy than in example 2 because of the stiffer section!

### Example 4

The final example in this section is a cranked beam problem, typical of many that we come across in real structures. Calculate the total elastic strain energy in this case.

Viz:



| Region | Moment Eqn.  | Range (m)        |
|--------|--------------|------------------|
| AB     | $+Ws$        | $s = 0, s = 0.4$ |
| BC     | $+0.4W$      | $y = 0, y = 0.5$ |
| CD     | $+0.4W - Wz$ | $z = 0, z = 0.9$ |

(ignoring direct strain energy)

$$U = \int_0^L \frac{M^2}{2EI} dx = \int_A^B \frac{M^2}{2EI} ds + \int_B^C \frac{M^2}{2EI} dy + \int_C^D \frac{M^2}{2EI} dz$$

$$U = \frac{1}{2EI} \int_0^{0.4} (Ws)^2 ds + \frac{1}{2EI} \int_0^{0.5} (0.4W)^2 dy + \frac{1}{2EI} \int_0^{0.9} (0.4W - Wz)^2 dz$$

$$U = \frac{W^2}{2EI} \left\{ \left[ \frac{s^3}{3} \right]_0^{0.4} + [0.16y]_0^{0.5} + \left[ \frac{z^3}{3} - \frac{0.8z^2}{2} + 0.16z \right]_0^{0.9} \right\}$$

$$\therefore U = \frac{W^2 \times 10^{-3}}{2EI} (164.33) Nm$$

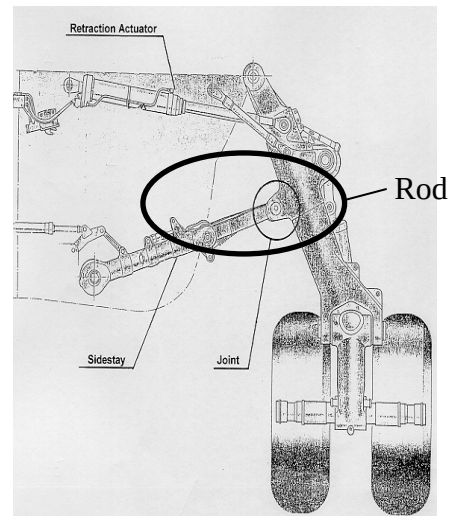
## Tutorials

### Question 1

A component in an undercarriage can be considered as a constant section rod under direct axial loading and pinned at either end. The rod is 0.5m long with a constant diameter of 30mm.

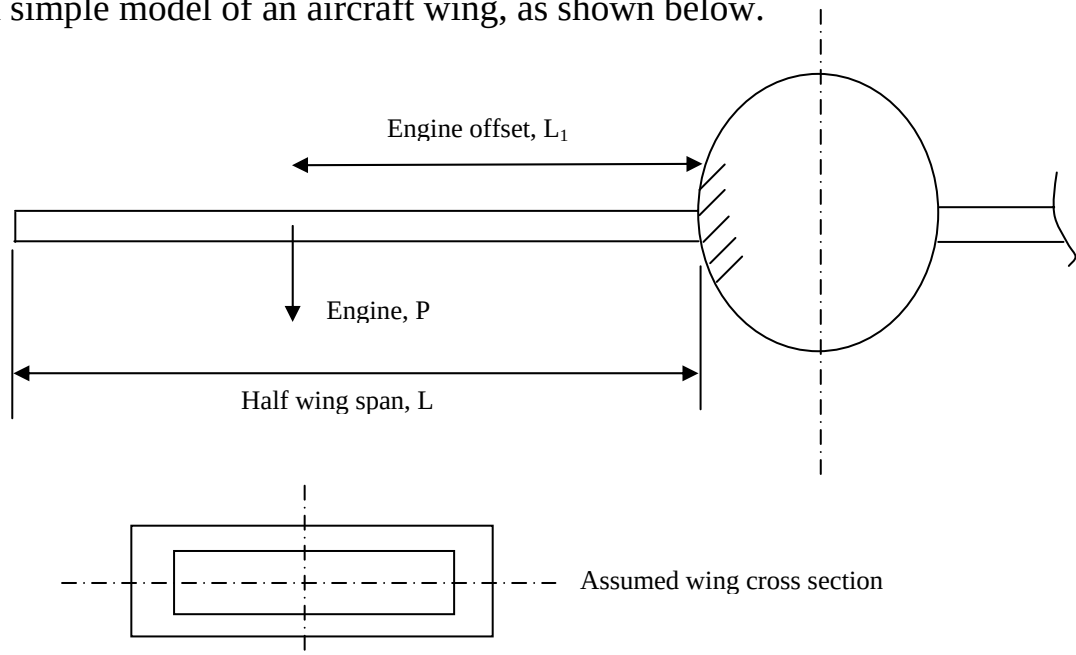
The rod is made from steel with a Young's Modulus value of  $210\text{GN/m}^2$ .

Determine the total strain energy stored in the rod when subjected to an axial tensile load of 1kN. What is the value if a compressive load of the same magnitude is now applied?



## Question 2

Consider a simple model of an aircraft wing, as shown below.



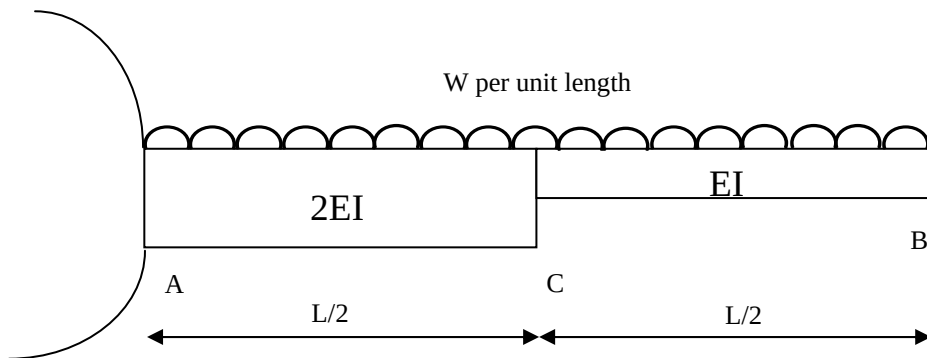
The fuselage can be considered as a fixed end constraint.

Assume values for  $L$ ,  $L_1$ ,  $P$  and cross-sectional dimensions and then calculate the total energy stored in the wing.

Assume the only significant load is the point load due to the engine.

**Question 3**

The wing of an aircraft can be modelled as a cantilever, with a step change in section, as shown below. The wing has a total length  $L$ . Section AC has a bending stiffness of  $2EI$  and section CB a bending stiffness of  $EI$ . Assuming the lift to be uniformly distributed along the wing, and equal to  $W$  per unit length, calculate the total strain energy stored due to bending.





## Energy Methods

There are for our purposes two ways of using energy to solve problems found in engineering 1) simple exchange of energy and 2) Unit Load method.

### Simple Exchange of Energy

The first approach simply equates external work done to internal strain energy. This is a simple and direct method but one that is very restrictive in its application.

Considering again Introduction to Strain Energy - Example 1, and now using the simple exchange of energy method, we can calculate the deflection of the simple framework example. Suppose that we want to find the vertical deflection at A.

External work done by the 20kN force as it moves downwards from '0' to  $\delta_{\max}$  as force increases,

$$\frac{20 \times 10^3 \cdot \frac{\delta_{\max}}{2}}$$

Equating energy,

$$\frac{20 \times 10^3 \cdot \frac{\delta_{\max}}{2}}{2} = 0.844$$

$$\therefore \delta_{\max} = \frac{0.844 \times 2}{20 \times 10^3} = 0.084 \times 10^{-3} \text{ m}$$

Considering again Introduction to Strain Energy - Example 2, and using the simple exchange of energy method, we can calculate the tip deflection of the simple cantilever. If vertical deflection at the tip is  $\delta_{\max}$  with the 20kN load fully applied,

$$\therefore \delta_{\max} = \frac{10.70 \times 10^3 \times 2}{20 \times 10^3} = 1.07 \text{ m} !!$$

This should warn us that the beam is in fact overloaded!

Check with the standard solution, using  $\delta = \frac{wl^3}{3EI} = 1.072 \text{ m}$  Check!

Now considering again Introduction to Strain Energy - Example 3, and using the simple exchange of energy method, we can calculate the tip deflection of the stepped cantilever. If vertical deflection at the tip is  $\delta_{\max}$  with the 20kN load fully applied,

$$\therefore \delta_{\max} = \frac{2.89 \times 10^3 \times 2}{20 \times 10^3} = 0.289m \quad \text{Better!}$$

Finally, considering again Introduction to Strain Energy - Example 4, and using the simple exchange of energy method, we can calculate the horizontal deflection at 'A' of the cranked beam. If the horizontal deflection at 'A' is  $\delta_{\max}$  with the load of W fully applied,

$$\therefore \delta_{\max} = \frac{164.33 \times 10^{-3} \cdot W}{EI} \text{ m}$$

## The Unit Load Method

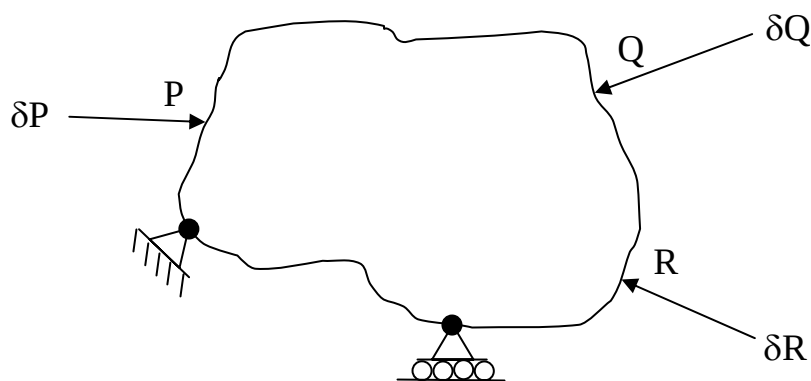
The second approach is more interesting as it can be used to predict any deflection. Unfortunately, as problems become more complicated it does start to involve lots of algebra.

The Unit Load method was developed out of a concept introduced by Carlos Alberto Pio Castigliano (1847-1884). The method of Virtual Work is similar.

Let  $U$  be the total energy in a system due to forces  $P, Q, R, \dots$  etc.

The system being in a state of equilibrium.

The reactions do NOT do work if they are rigid!



If  $P, Q, R, \dots$  change by amount  $\delta P, \delta Q, \delta R, \dots$

Then from traditional maths concerns with 'error' or small changes we know that if 'A' is a quantity and  $x, y$  and  $z$  control its value:

$$\text{Eg. } A = 3x^2 + 7y + \sin(4z)$$

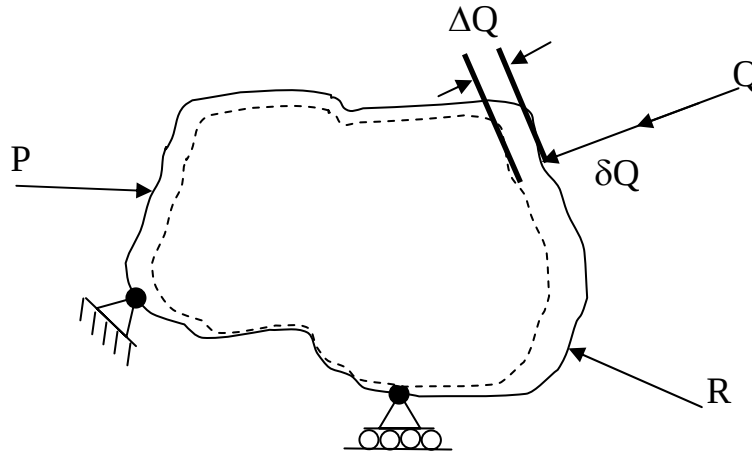
$$\delta A = \frac{\partial A}{\partial x} \cdot \delta x + \frac{\partial A}{\partial y} \cdot \delta y + \frac{\partial A}{\partial z} \delta z$$

In our case:

$$\delta U = \frac{\partial U}{\partial P} \cdot \delta P + \frac{\partial U}{\partial Q} \cdot \delta Q + \frac{\partial U}{\partial R} \delta R + \dots$$

If only 'Q' changes then  $\delta U = \frac{\partial U}{\partial Q} \cdot \delta Q$

Consider now the same elastic system but with  $\delta Q$  alone acting. If we then apply P, Q and R the system will deform and in particular Q will displace by amount  $\Delta Q$ .



Therefore, force  $\delta Q$  will travel distance  $\Delta Q$  and do work  $\delta Q \cdot \Delta Q$

$$\text{Therefore, } \delta U = \delta Q \cdot \Delta Q$$

Since the energy total is the same whatever the sequence of loading we can equate our two expressions for  $\delta U$ .

$$\frac{\partial U}{\partial Q} \cdot \delta Q = \delta Q \cdot \Delta Q$$

$$\therefore \Delta Q = \frac{\partial U}{\partial Q}$$

In words: If an elastic system is loaded with forces P, Q, R etc. the deformation at Q, in the direction of Q, is given by the partial derivative of the total energy expression with respect to Q

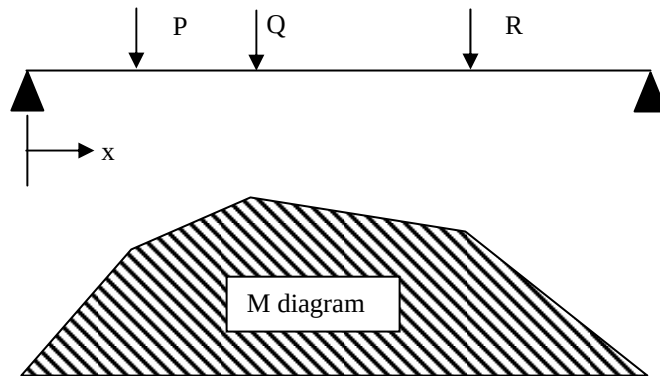
As we want a completely general purpose and practical approach it will pay us to look at  $\Delta Q = \frac{\partial U}{\partial Q}$  in more detail, before seriously using it!

For **bending** we know that energy is given by:  $U = \int_0^L \frac{M^2}{2EI} dx$

We can therefore write:

$$\therefore \Delta Q = \frac{\partial U}{\partial Q} = \frac{\partial}{\partial Q} \left( \int_0^L \frac{M^2}{2EI} dx \right)$$

If forces P, Q, R are present then M will be a function of P, Q, R ....



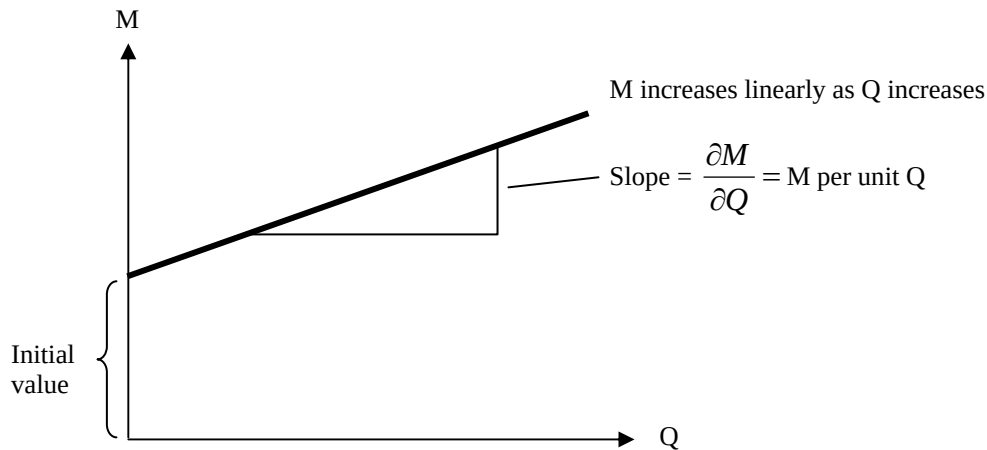
If we are careful we can differentiate first and then integrate because the differential is with respect to force and the integral is with respect to length.

Using the 'Function of a Function' rule  $\frac{\partial U}{\partial Q} = \frac{\partial U}{\partial M} \cdot \frac{\partial M}{\partial Q}$  we can write:

$$\Delta Q = \int_0^L \frac{2M \left( \frac{\partial M}{\partial Q} \right)}{2EI} dx$$

If Q alone were allowed to change then the moment value at ANY point will increase linearly.

Hence, at any point 'x' along the beam, the value of bending moment M will increase linearly as Q increases:



The significance of this is that we need never perform the differential as mathematically we can substitute a moment (small 'm') expression for  $\frac{\partial M}{\partial Q}$  based on  $Q=1$ .

This is the basis of the 'Unit Load Method' and can be written as:

$$\Delta Q = \int_0^L \frac{M \cdot m}{EI} dx$$

Where:

M is due to REAL loading

m is due to UNIT loading at place where the deflection is required

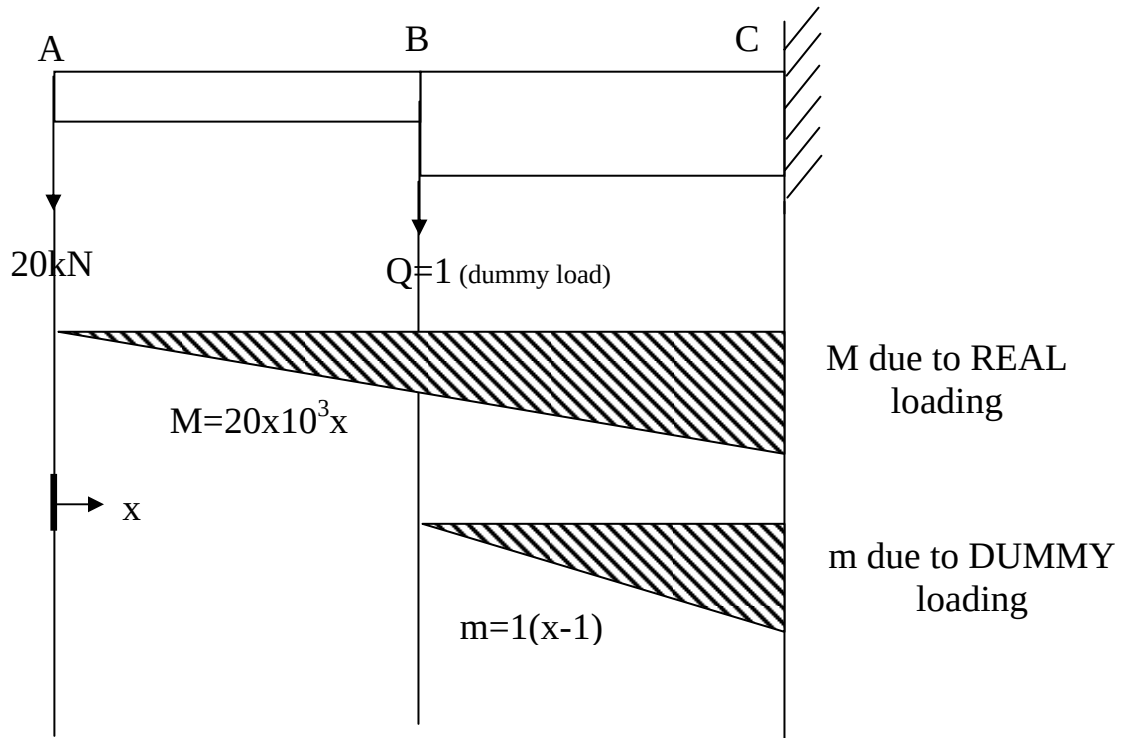
It does not matter whether there is a REAL load at the deflection point or not!

We are now in a position to cope with almost any system of loading and any geometry (shape) of the beam.

### Example 1

Now considering again Introduction to Strain Energy - Example 3, and using the Unit Load method, we can calculate the deflection at point B on the stepped cantilever.

VIZ:



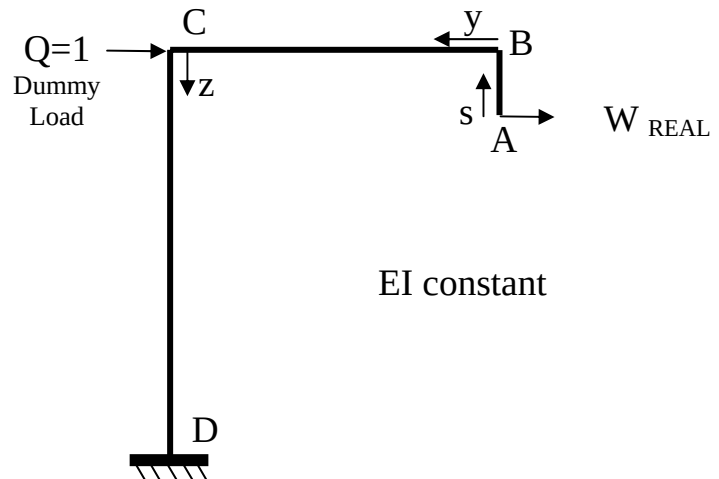
| Region | M (Nm)             | m (Nm)   | M.m                        | I (m <sup>4</sup> )    |
|--------|--------------------|----------|----------------------------|------------------------|
| AB     | $20 \times 10^3 x$ | 0        | 0                          | $0.712 \times 10^{-6}$ |
| BC     | $20 \times 10^3 x$ | $1(x-1)$ | $20 \times 10^3 (x^2 - x)$ | $4.288 \times 10^{-6}$ |

To find the deflection at B:

$$\begin{aligned}
 \Delta Q &= \int_0^L \frac{M \cdot m}{EI} dx = \int_0^1 \frac{Mm}{EI} dx + \int_1^2 \frac{Mm}{EI} dx \\
 &= 0 + \int_1^2 \frac{20 \times 10^3 (x^2 - x) dx}{70 \times 10^9 \cdot 4.288 \times 10^{-6}} \\
 &= 55.5 \times 10^{-3} \text{ m}
 \end{aligned}$$

## Example 2

Now considering again Introduction to Strain Energy - Example 4, and using the Unit Load method, we can calculate the horizontal deflection at point C on the cranked beam.



| Region | M (Nm)      | m (Nm) | M.m             |
|--------|-------------|--------|-----------------|
| CD     | $0.4W - Wz$ | $-1.z$ | $W(z^2 - 0.4z)$ |

$$\begin{aligned}
 \Delta Q &= \int_0^L \frac{M \cdot m}{EI} dx = \int_C^D \frac{M \cdot m}{EI} dz = \int_{z=0}^{z=0.9} \frac{W(z^2 - 0.4z)}{EI} dz \\
 &= \frac{W}{EI} \left[ \frac{z^3}{3} - \frac{0.4z^2}{2} \right]_0^{0.9} \\
 &= \frac{W}{EI} \left\{ \left[ \frac{0.9^3}{3} - \frac{0.4 \times 0.9^2}{2} \right] - 0 \right\} \\
 &= \frac{W}{EI} \{0.243 - 0.162\} \\
 &= 0.081 \frac{W}{EI}
 \end{aligned}$$

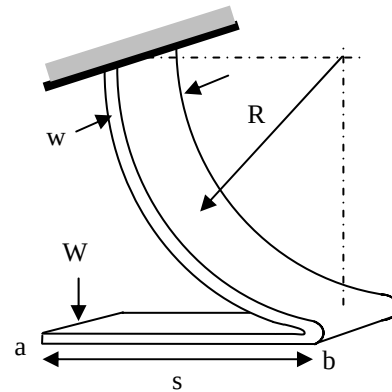


The Unit Load method can also be developed further to deal with more complex beam shapes, including curved beams.

### Example 3

The flex-beam shown below is designed in such a way that a force  $W$  applied vertically at 'a' causes horizontal movement only at 'b'.

Determine the necessary length of dimension 's' to ensure zero vertical deflection at 'b' due to 'W'. Give your answer in terms of 'R'. Assume 't' is small compared with 'R' and that 'bc' is a quadrant of a circle fixed at 'c'.



Using the UNIT LOAD method

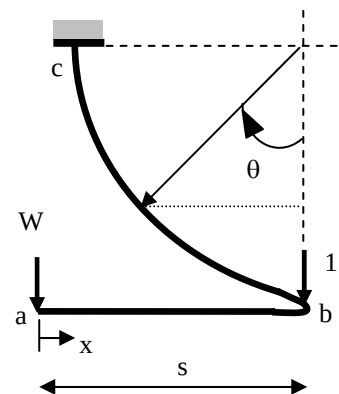
$$\Delta = \int \frac{Mm}{EI} ds$$

Apply a Unit Load at b, as shown:

$$\Delta_b = \int_a^b \frac{Mm}{EI} dx + \int_b^c \frac{Mm}{EI} R d\theta$$

For region ab:  $M = Wx$   
 $m = 0$

For region bc:  $M = W(s - R\sin\theta)$   
 $m = -1 \times R\sin\theta$



$$\Delta_b = \int_a^b \frac{Mm}{EI} dx + \int_0^{\pi/2} \frac{W}{EI} (s - R\sin\theta) \cdot (-R\sin\theta) \cdot R d\theta$$

But the vertical deflection at b is zero (a given condition).

$$0 = -\frac{WR^2}{EI} \int_0^{\pi/2} (s - R \sin \theta) \sin \theta d\theta$$

$$= \int_0^{\pi/2} (s \sin \theta - R \sin^2 \theta) d\theta$$

$$= \int_0^{\pi/2} \left[ s \sin \theta - R \left( \frac{1 - \cos 2\theta}{2} \right) \right] d\theta$$

$$= \left[ -s \cos \theta - \frac{R}{2} \left( \theta - \frac{1}{2} \sin 2\theta \right) \right]_0^{\pi/2}$$

$$= \left[ -0 - \frac{R}{2} \left( \frac{\pi}{2} - 0 \right) \right] - \left[ -s - 0 \right]$$

$$\therefore 0 = -R \frac{\pi}{4} + s$$

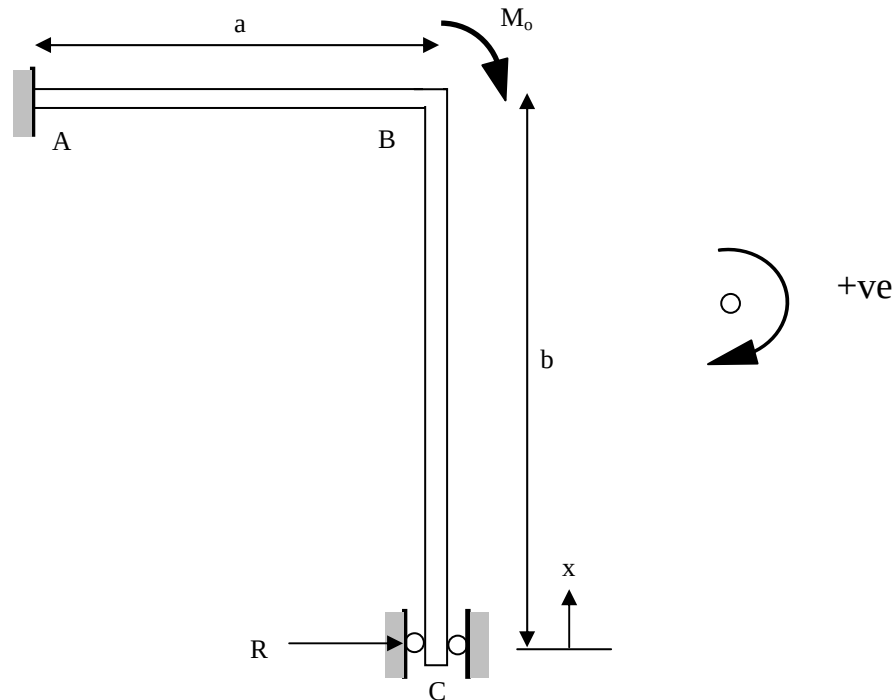
$$\therefore s = \frac{\pi R}{4}$$

---

For zero deflection at b.

### Example 4

A plane frame is loaded by a moment  $M_0$  at B. Determine the reaction force 'R' at the roller, by application of the Unit Load Method.



For segment BC:  $M_1 = -Rx$        $m_1 = -x$

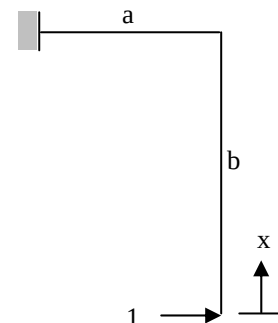
For segment AB:  $M_2 = -Rb + M_0$        $m_2 = -b$

$$\delta_c = 0 = \int_0^b \frac{M_1 m_1}{EI} dx + \int_0^a \frac{M_2 m_2}{EI} dx$$

or

$$0 = \int_0^b Rx^2 dx + \int_0^a (-Rb + M_0)(-b) dx$$

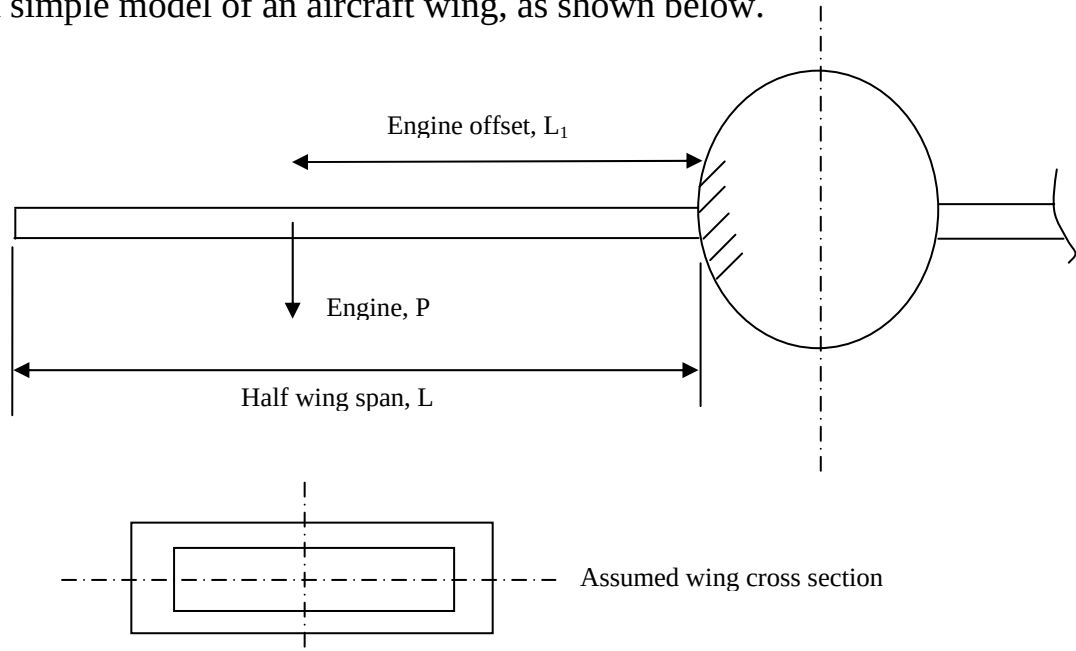
$$0 = \frac{Rb^3}{3} + Rb^2a - M_0.ab \quad \therefore R = \frac{3M_0.a}{b(b+3a)}$$



## Tutorials

### Question 1

Consider a simple model of an aircraft wing, as shown below.



The fuselage can be considered as a fixed end constraint.

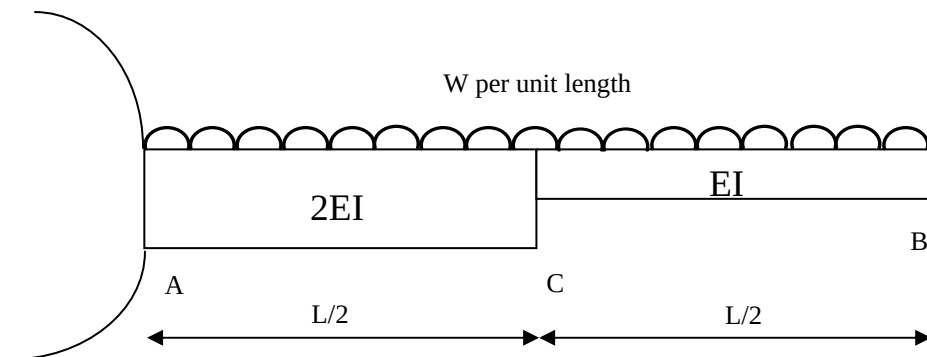
Assume values for  $L$ ,  $L_1$ ,  $P$  and cross-sectional dimensions and also assume the only significant load acting is the point load due to the engine.

- Make an initial estimation of the wing tip deflection.
- Once completed, repeat your calculation adding in a continuous (or discontinuous) UDL, in addition to the engine load. Calculate the new wing tip deflection and slope.
- Once completed, repeat your calculation but this time model your wing as a tapered beam, starting at the above chosen cross section at the wing root and tapering to nothing at the wing tip. In other words define  $I$  as a function of  $x$  and then integrate it! Calculate the new wing tip deflection.

## Question 2

The wing of an aircraft can be modelled as a cantilever, with a step change in section, as shown below. The wing has a total length  $L$ . Section AC has a bending stiffness of  $2EI$  and section CB a bending stiffness of  $EI$ . Assuming the lift to be uniformly distributed along the wing, and equal to  $W$  per unit length, calculate:

- the deflection at the mid-span, C;
- the deflection at the wing tip, B.



### Question 3

An aeroplane weighing 50kN has wings cantilevered from the fuselage. Each wing is 5m long and carries an engine weighing 2kN located at 2.5m from the wing's root.

The wing structure can be assumed to be a uniform cantilever beam for which the second moment of area,  $I = 104.10^{-6} \text{ m}^4$  and Young's Modulus,  $E$  for is  $70.10^6 \text{ kN/m}^2$  and has a weight of 30kN (for both wings). Assuming the lift to be uniformly distributed along the wings calculate, using the UNIT LOAD method, the rise of the wing tips during flight from the horizontal position.

In all calculations ignore the tail lift and assume the lift of the aircraft equals the total weight and is applied as shown below.

